

Karthikeyan Devadoss

Forward Deployed Engineer (AI infrastructure) | Backend & cloud foundation

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SUMMARY

Applying for Forward Deployed Engineer, Physical AI Infrastructure at Nebius. Engineer focused on AI solution architecture, with about ten years of backend development since December 2016. Since early 2026, one of a handful of engineers in NRG Energy's initial core AI team, designing and building AI solutions hands-on alongside senior backend work, including production AI triage with a human approval step, adopted by several teams. 4+ years at NRG combining hands-on Java/Spring development with architecture discussions and design decisions alongside the lead. Independently designed and built a public agentic software-delivery platform: approval-gated agents, a deterministic vs LLM decision split, RAG with labelled retrieval evals and MCP tool boundaries.

CORE SKILLS

Backend	Java, Spring Boot, REST, GraphQL, JWT/FusionAuth, AWS Lambda/SAM, SQS, Oracle, Docker, Jenkins, JUnit, Kafka, FHIR, Apache Camel
AI architecture	AI solution design · human-approval & capability boundaries · deterministic vs LLM decision split · RAG & retrieval evaluation (recall/MRR) · MCP & agent tool boundaries
AI stack	Azure OpenAI, vector retrieval (production, NRG) · Python, Anthropic API, embeddings, Postgres, Playwright (personal project)

EXPERIENCE

Senior Backend Developer — NRG Energy

End client; employed via Remote GmbH

Mar 2022 – present

- Architecture:** took part in architecture discussions and design decisions with the lead on a multi-million-customer retail-energy platform.
- Initial core AI team (since early 2026):** one of a handful of engineers in the team, designing and building AI solutions hands-on alongside senior backend work.
- Production AI triage:** Azure OpenAI with vector retrieval and a human email-approval boundary before any action is taken; adopted by several teams; ran informal teaching sessions.
- Hands-on Java/Spring Boot: REST, GraphQL, JWT/FusionAuth; 2025 AWS Lambda/SAM; SQS with dead-letter queues.
- Mentored offshore developers (sprint breakdown, guidance, PR review); monthly production on-call.

Java Developer Consultant — Blue Cross Blue Shield Association (BCBSA)

Jun 2019 – Mar 2022

- Java/Spring Boot REST services for healthcare integration; hands-on FHIR in member/coverage/claims interoperability.
- Application-side Kafka producers and consumers for domain events; careful handling of sensitive healthcare data.

Java Backend Developer — Marsh (Digital Broker / Bluestream)

Via Innovecture LLC

Dec 2016 – Jun 2019

- Java/Spring APIs and carrier integration (Apache Camel routes, ACORD-oriented mapping), implemented within the technical architect's design.

PERSONAL PROJECT

Agentic Software Delivery Platform

PERSONAL PROJECT

Independently designed and built

Live demo · GitHub repo

- Deterministic vs LLM split:** deterministic checks kept separate from LLM proposals; a Python agent (Anthropic API) plans and applies changes to a real Spring Boot target via tool calling.
- Human-approval boundary:** structural approval gate; the model has no code path to approve its own change, and approval is bound by content hash to the exact change.
- RAG eval design:** local retrieval (embeddings, cosine similarity, content-hash indexing) with a labelled eval (recall/MRR).
- MCP / tool boundaries:** MCP server for read-only repository tools; independent QA subagent without write tools; durable Postgres event ledger with versioned pricing.

EDUCATION

B.E. Computer Science — Madras University, Chennai

LANGUAGES

English — fluent · German — A1-A2